Xi'an Jiaotong-Liverpool University
西交利物浦大学

Department of Biological Sciences

BIO204 Advanced Genetics - Lab Report 3

Name of Practical: Reverse Genetics Analysis of AN8455

Author: Yuewei Xu

	Student 1	**Student 2**	**Student 3**
Peer Mark by Student 1		100%	100%
Peer Mark by Student 2	100%		100%
Peer Mark by Student 3	100%	100%	
Relative Peer Contribution	100%	100%	100%
Group Lab Report Mark			
Individual Lab Report Mark			

Reverse Genetics Analysis of AN8455 in *Aspergillus nidulans* Through Af-pyrG+ Knockout-Cassette Mediated Gene Replacement

1. Abstract

Reverse genetics enables gene function investigation by a workflow from genotype to phenotype. This study applied an Af-pyrG+ knockout-cassette mediated workflow to analyze the uncharacterised *Aspergillus nidulans* gene AN8455. Bioinformatics analysis identified AN8455 as a conserved but functionally unresolved fungal gene predicted to encode an intracellular, non-secreted, and non-transmembrane protein containing a DUF3405 domain, and further defined the native AN8455 genomic locus, the expected Af-pyrG+ replacement structure, and the diagnostic PCR products for both strains. Following transformation, putative transformants were isolated and purified, genomic DNA was extracted, and diagnostic PCR was performed to test gene replacement. Agarose gel electrophoresis photos showed that the original strain control yielded a 1195 bp band, close to the predicted 1124 bp product (+6.93%), whereas the two putative transformants yielded 1044 bp and 1021 bp bands, both consistent with the expected 1023 bp transformant product (+2.05% and −0.20%), supporting the successful replacement of the endogenous AN8455 ORF by the Af-pyrG+ cassette. Additional bands detected in the AN8455-F/AN8455-R reactions of transformant samples were further analyzed and interpreted as non-specific amplification under improper PCR conditions. Overall, the study validated the design-to-genotype section of reverse-genetic analysis for AN8455 and established a basis for the subsequent phenotype-based functional characterisation.

2. Introduction

2.1 Reverse genetics and knockout cassette

In genomic researches, the main strategy has shifted from classical forward genetics to reverse genetics (1). While classical forward genetics identifies gene function by examining an altered phenotype after random or induced mutation (2), reverse genetics tests functionally changed genes through phenotypic analysis after targeted gene disruption (3). Reverse genetics therefore works from gene to phenotype and enables targeted functional analysis of genes whose sequences are already known but biological roles remain unclear. This shift has become especially important in the post-genomic era, because rapid development in sequencing technology has generated far more known gene sequences than functionally characterised genes (1).

One common method to perturbate the target gene in reverse-genetics experiments is targeted replacement of the gene with a synthetic DNA construct called a knockout cassette. A knockout cassette is a gene-specific linear DNA fragment designed to mediate targeted deletion through homologous recombination (4). It typically contains three components: a selectable marker gene in the middle, flanked by two homologous DNA arms identical to the upstream and downstream sequences of the target gene in the genome (5). Through homologous recombination, the functional endogenous gene is replaced by the marker sequence, thereby deactivating the original gene. This gene disruption allows the resulting defects to be observed and used to infer the original function of the target gene.

2.2 *Aspergillus nidulans*

Aspergillus nidulans (*A. nidulans*) is a filamentous fungus that is non-pathogenic but closely related to several plant and animal pathogens (6). It has long served as a model organism for studies of eukaryotic gene function because it combines a moderately-sized genome, which is already well annotated, a substantial number of protein-coding genes, a stable haploid life cycle, and the capacity to produce sexual spores (7-10).

2.3 Af-pryG+, ΔnkuA and AN8455

In this study, an *A. nidulans* pyrG89 ΔnkuA pyrA4 strain carrying the AN8455 locus was used as the host for transformation, together with a knockout cassette containing the selectable marker Af-pyrG+, to establish the complete transformation system.

The gene of interest was AN8455 (ANID_08455), an uncharacterised protein-coding gene located on the negative strand of chromosome V (541006–543575) in the *A. nidulans* FGSC A4 genome. Orthologs are present in several Aspergillus species, indicating conservation within this lineage. AN8455 contains eight exons and has a single annotated transcript, AN8455-T. Transcriptomic data suggest condition-dependent regulation, including differential expression across nitrogen sources, developmental stages, and selected regulatory mutant backgrounds. The gene encodes a 653-amino-acid protein with a predicted molecular weight of 75,391 Da and an isoelectric point of 5.75. Bioinformatics analysis predicted that this protein is intracellular, non-secreted, and lacks both transmembrane helices and a signal peptide. Although no specific molecular function has yet been assigned, the protein contains a highly conserved DUF3405 domain (PF11885; amino acids 143–640) and is most likely localized to the cytoplasm or nucleus (FungiDB–(11), Appendix I).

The selectable marker Af-pyrG+ derived from Aspergillus fumigatus. It encodes orotidine 5′-phosphate decarboxylase, the terminal enzyme in uridine 5′-phosphate biosynthesis (12). As the host strain carries a loss-of-function Af-pyrG+ mutation (pryG89), it cannot grow in the absence of exogenous uridine and uracil. Only cells that successfully integrate the knockout cassette regain growth on selective medium, allowing specific recovery of transformants and representing principle of auxotrophic selection system in this study.

To further facilitate targeted integration instead of random insertion, the host strain also carries a ΔnkuA mutation. The nkuA gene is the fungal orthologue of human Ku70, a core component of the non-homologous end joining (NHEJ) pathway (13). Deletion of nkuA suppresses NHEJ and shifts DNA repair towards homologous recombination after transformation induced double strand breaks (13). This substantially increases the frequency of targeted gene replacement, in some cases from below 5% to above 90% (13).

2.4 Aim of study

This study aimed to characterize the gene AN8455 in *A. nidulans* by applying a reverse-genetics workflow that combines bioinformatics analysis, knockout-cassette design, gene transformation, candidate purification, and diagnostic PCR validation, while leaving phenotype analysis as a future plan, and trying to investigate any special observations during the process for further improvement.

3. Materials & Methods

3.1 Knockout cassette assembly and amplification

The AN8455 knockout cassette was pre-designed and assembled to replace the endogenous AN8455 open reading frame (ORF) with the selectable marker Af-pyrG+. The cassette contained the 5′ and 3′ flanking regions of AN8455, which were PCR-amplified using primer pairs outside the AN8455 ORF (5′F/5′R and 3′F/3′R; Appendix I, II). The left side of the 5′ flank included the AN8455-Forward sequence (Appendix I, II), which was later used as a diagnostic primer. In addition, the 3′ end of the 5′ flank and the 5′ end of the 3′ flank carried the CDS164 and CDS165 sequences (Appendix III), respectively, to enable recombination with the Af-pyrG+ fragment during cassette construction. Notably, part of CDS164 also corresponded to the diagnostic primer Af-pyrG-Reverse. The separately amplified 5′ flank, 3′ flank, and Af-pyrG+ were then inserted into a plasmid, whose assembly was completed by yeast-mediated recombination (Figure 1).

After assembly, the knockout cassette was further amplified by PCR to generate sufficient DNA for

fungal transformation. The PCR products were checked by agarose gel electrophoresis. Cassettes with a validated band of approximately 2.5 kb were used in the subsequent transformation experiment.

3.2 Cassette transformation and putative transformants purification

Before transformation, fungal mycelia were treated using cell wall-degrading enzymes to generate protoplasts, and resuspended in transformation solution to prepare fragile protoplasts capable of DNA uptake.

For transformation, 100 µL of protoplast suspension was mixed with either 20 µL of knockout-cassette DNA or 20 µL of transformation solution as a negative control. Samples were treated with 50 µL PEG, incubated on ice for 20 min, supplemented with an additional 1 mL PEG, held at room temperature for 5 min, and then mixed with 5 mL transformation solution. The mixtures were plated onto selective regeneration medium by top-layer pouring with molten medium at approximately 60°C, using four plates for the transformation sample and two for the negative control. As the plates were a selective medium lacking exogenous uridine and uracil, only successful transformants were expected to survive and form colonies.

Three putative transformant colonies were then selected for purification, including one colony from the Group 11 transformation plate T2 (Figure 3B) and two colonies provided by the demonstrator. Each colony was transferred separately into a 1.5 mL tube containing 100 µL of sterile water and mixed for approximately 30 s to disperse the colony material. From each suspension, 10 µL was reseeded onto a fresh agar plate in three streaks to obtain cleaner and expanded colony growth for downstream analysis.

3.3 DNA extraction and diagnostic PCR

After one week of incubation, purified transformants from two re-streaked plates (P1 and P3, Figure 4B), together with an original untransformed strain provided by the demonstrator as a negative control (N1, Figure 4), were picked for genomic DNA extraction.

For each sample, hyphal material was lysed in 600 µL Buffer FG1 with 5 µL RNase (20 mg/mL) at 65°C for 10 min, followed by addition of 140 µL Buffer FG2 and incubation on ice for 5 min. After centrifugation (>10,000 × g, 10 min), the supernatant was mixed with 0.5 volume Buffer FG3 and 1 volume absolute ethanol, then applied to a DNA mini-column. The column was washed twice with 600 µL DNA wash buffer, dried by centrifugation, and genomic DNA was eluted with 100 µL preheated elution buffer (70°C) after 3–5 min incubation at room temperature. The eluate was used directly for PCR.

Diagnostic PCR was performed to determine whether the native AN8455 locus had been replaced by the Af-pyrG+ cassette. Two primer combinations were used: AN8455-For + AN8455-Rev (Appendix I) and AN8455-For + Af-pyrG-Rev (Appendix I). The PCR assay included four DNA sources: a water negative control, the original strain control, transformant 1, and transformant 2. Each DNA source was tested with both primer sets, giving a total of eight reactions. Each 20 µL reaction contained 15.2 µL PCR master mix, 4 µL DNA sample, and 0.8 µL mixed primer set. PCR products were analyzed by agarose gel electrophoresis.

3.4 Bioinformatics analyses

Bioinformatics analyses were conducted in three stages.

First, FungiDB (11) and ApE (Version 3.1.9) were used to define the genomic structure of AN8455 in the original strain. Gene location and strand orientation were retrieved from FungiDB. An additional 1200 bp was included on both the 5′ and 3′ sides to cover the upstream and downstream regions, and the flanking regions defined by the cassette-construction primers, together with the ORF between them, were annotated in ApE (Appendix II, Figure 1A). The AN8455 ORF

was then replaced in silico with the Af-pyrG+ sequence to generate the predicted transformant structure while retaining the same flanking regions (Appendix III, Figure 1B). Diagnostic primers were mapped onto both the original-strain model and the predicted transformant model, and the expected PCR product sizes were calculated: 1124 bp for AN8455-For + AN8455-Rev, and 1023 bp for AN8455-For + Af-pyrG-Rev (Appendix I, II, III).

Second, FungiDB was used for functional and structural annotation of AN8455. Retrieved information included genomic features, orthologues, transcript and exon organization, transcriptomic data, GO annotation, predicted protein sequence, and general protein properties (Appendix I). In addition, both the protein encoded by AN845 and Af-pyrG+ were submitted to AlphaFold 3 for structural prediction, and the resulting model showed overall high confidence scores (Appendix I).

Third, experimental outputs were analyzed quantitatively. For diagnostic PCR gels, migration distances of DNA ladder bands and sample bands were measured in ImageJ (Version p1.54), and were then analyzed in R (Version 4.4.3), where a log-linear calibration model was fitted using the ladder bands. To investigate unexpected bands, putative non-specific primer annealing sites were also analyzed, including GC content, local sequence alignment, and predicted melting temperature (Tm) using ApE and R.

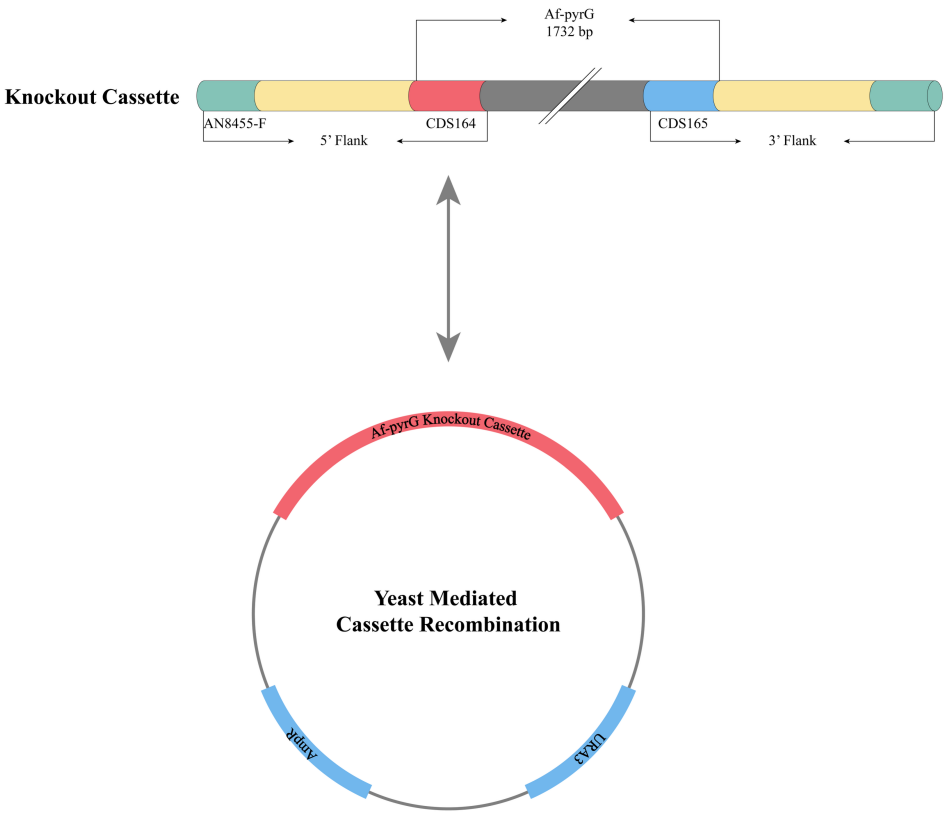


Figure 1. *Construction of the Af-pyrG knockout cassette and its assembly by yeast-mediated recombination. The Af-pyrG knockout cassette is composed of 5'flank, 3'flank and the Af-pyrG+ selectable marker. The cassette assembly was completed by yeast-mediated recombination in the plasmid containing AmpR and URA3.*

4. Results & Analyses

4.1 Genomic organization of the AN8455 locus and predicted Af-pyrG replacement structure

The structure of the AN8455 locus was reconstructed using retrieved, resulting in a 4970 bp region located on the negative strand of chromosome V (coordinates 541006–543575, extended ±1200 bp) (Figure 2A, Appendix II). Within this region, the AN8455 ORF spans 2519 bp (positions 1211–3729). The flanking regions were defined by primer positions: the 5′ flank (877 bp) is bounded by primers 5'F and 5'R, while the 3′ flank (690 bp) is defined by primers 3'F and 3'R. For diagnostic primers, AN8455-F is located upstream of the 5′ flank (position 209), while AN8455-R lies within the ORF (ending at position 1332). Therefore, amplification using AN8455-F + AN8455-R yields a 1124 bp product only when the endogenous ORF is present.

The knockout strategy replaces the AN8455 ORF with the Af-pyrG selectable marker sequence (1732 bp) (Figure 2B, Appendix III). The recombination junctions are defined by CDS164 and CDS165, which are incorporated into the cassette as overlap regions. Notably, the 5′ portion of CDS164 also serves as the Af-pyrG-R diagnostic primer binding site. As a result, amplification using AN8455-F + Af-pyrG-R produces a 1023 bp product only in transformants carrying the correct insertion.

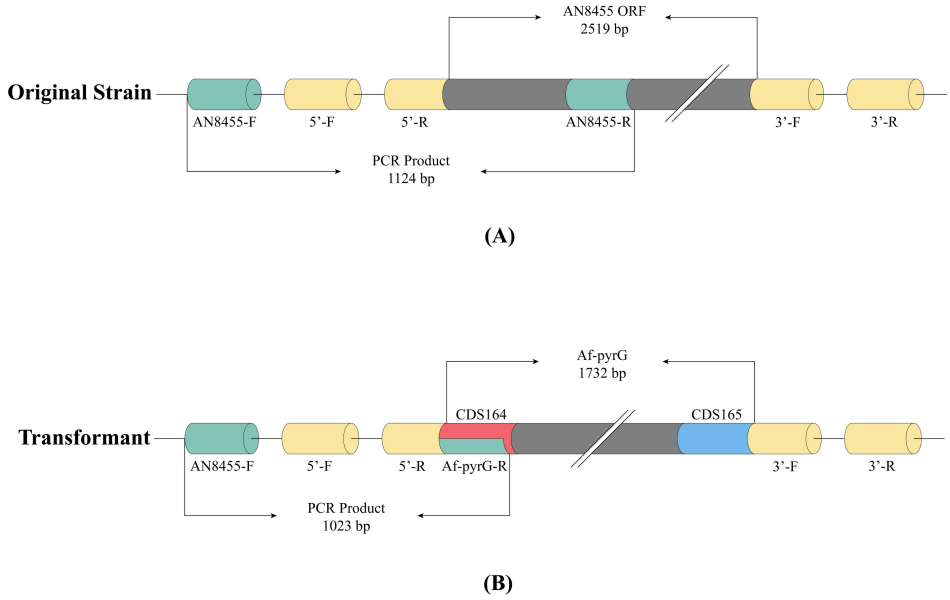


***Figure 2**. Structure illustrations of AN8455 locus in original type and Af-pyrG+ site in transformant. **(A)** For the original strain, AN8455 ORF is 2519 bp, with an expected PCR product of 1124 bp. **(B)** For transformant, Af-pyrG+ replacement is 1732 bp, with an expected PCR product of 1023 bp.*

4.2 Transformation and purification of putative transformants

Transformation in Group 35 yielded no usable colonies. No visible colonies appeared either in control groups (C1–C2) or treatment plates (T1–T4). Therefore, the transformation for Group 35 was considered unsuccessful, and possible explanations for this failure are addressed in the discussion section.

Because of this failure, subsequent analysis was based on Group 11 plates, which showed clear differential growth between control and treatment conditions

(Figure 3A, B). Control plates (C1–C2) were almost blank, with only a few small, non-filamentous brown specks, lacking radial expansion and therefore interpreted as background, and supported our design that only successful transformant can grow in plates without uridine and uracil. In contrast, the treatment plates (T1, T2, and T4) showed colony formation with fungal morphology, indicating successful recovery under selection, although colony quality differed across plates. T1 showed only minimal colony growth. The two tiny colony-like, faint yellow-green residuals lack clear radial expansion or conidiation and were therefore not considered reliable. T2 contained a single dominant colony that was clearly distinguishable from background. This colony showed broad radial expansion from a dense light yellow-green central inoculation point, consistent with active conidiation. A surrounding zone of filamentous peripheral growth transitioning from light green to white. The colony was clearly diffuse and filamentous, indicating not only vigorous hyphal extension but also a coherent, well-developed, and consistent pattern with expected *A. nidulans* growth, making it the most reliable candidate for downstream purification. T3 showed no clear colonies. T4 exhibited obvious heterogeneous growth. Three visually distinct colony types were observed: the upper colony was compact and rounded with a yellow-brown center and pale margin; the middle formed a diffuse, irregular yellow-green patch with poorly defined edges; and the lower was a dark, almost black colony partially intersecting with the middle colony. These growth patterns differed substantially in color, density, and margin structure, and none exhibited clear radial expansion or dense aerial hyphae, suggesting the absence of a single clean transformant morphology. In addition, the irregular distribution and mixed pigmentation raised concerns about contamination, mixed growth, or unstable recovery under selection. Although this morphological heterogeneity provides potentially informative variation for understanding *A. nidulans* growth characteristics and transformant behavior, it was not pursued further due to experimental objectives and resource limitations.

At this stage, colonies were considered putative transformants, as growth on selective medium alone cannot distinguish correct homologous replacement from ectopic integration or mixed populations. Therefore, purification was required before molecular validation. Colonies for purification were selected as follows: one demonstrator-provided original strain as the negative group (N1), three positive groups, including two demonstrator-provided transformants (P1 and P3), and one colony from Group 11 T2 (P2). These were re-streaked onto fresh plates to obtain purified growth (Figure 4). N1, P1, and P3 all produced consistent colonies across streak lines.

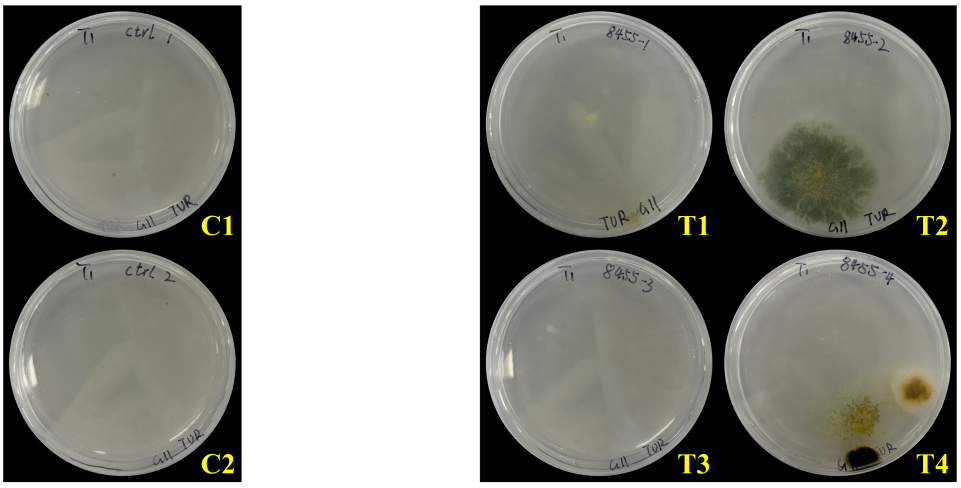
For each streak, colonies were clearly distinguishable from the background and exhibited a layered structure. At the base, colonies showed broad radial expansion from a central inoculation point. The central region appeared dark green, consistent with active conidiation, and was surrounded by peripheral mycelial growth that gradually transitioned from lighter green to pale yellow-white. The mycelium in this region appeared granular to short filamentous, and the colony overall was diffuse and filamentous, indicating vigorous hyphal extension. At the outermost edge, the colony transitioned into pale yellow-white, sparse peripheral aerial hyphae. Above the dark-green central region, an additional dense, elevated layer was present, composed of tightly packed white flocculent hyphae. During subsequent sampling, this upper layer could be physically removed and showed slight viscosity, allowing separation from the underlying structure. Together, these features indicate stable colony growth and are highly consistent with successfully established transformants. Compared with P1 and P3, colonies in N1 showed highly similar morphology, with the only difference being that the outermost pale yellow-white flocculent layer appeared slightly denser in color in N1 than in P1 and P3. In contrast, colonies in P2 showed markedly different characteristics. Across the three streak regions, very dark green, tar-like

aggregates were observed. These appeared as discontinuous, point-like deposits and did not exhibit clear radial expansion, although one region showed a concentrated dark-green center with visible colony formation. In addition, colonies resembling the typical transformant observed in other groups appeared randomly distributed across the plate in points rather than along streak lines. This growth pattern deviates substantially from the expected morphology of *A. nidulans*, suggesting possible contamination or unstable growth. Therefore, P2 was excluded from subsequent DNA extraction and diagnostic PCR analysis.

Overall, based on colony morphology, conidiation, and growth stability, N1, P1, and P3 were more consistent with expected purified original strain/transformant characteristics and were therefore selected for downstream molecular analysis.

expansion. Although occasional colony-like structures appeared, they did not exhibit clear filamentous growth or consistent organization. In addition, colonies resembling typical morphology appeared sporadically across the plate rather than along streak lines. This irregular distribution and inconsistency suggest contamination or unstable growth. Therefore, P2 was excluded from subsequent analysis.

Overall, based on colony morphology, conidiation, and growth stability, N1, P1, and P3 were considered reliable and were selected for downstream molecular validation.



(A) **(B)**

Figure 3. *Transformation outcome plates.* ***(A)*** *Control groups (C1–C2), no clear colonies, showing validity of environment and auxotroph design.* ***(B)*** *Treatment groups (T1–T4), showing rich morphological heterogeneity. Colony in T2 was picked as putative transformant, and purified to obtain P2 in Figure 4.*

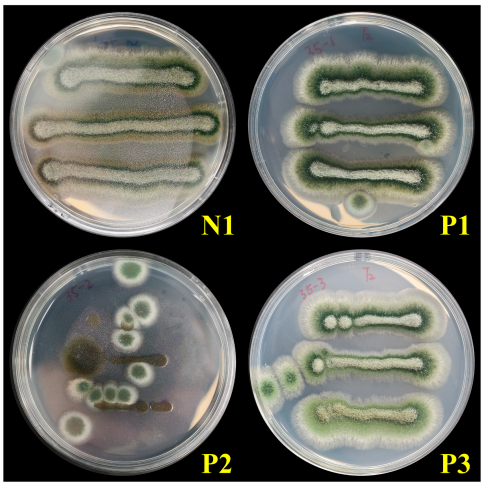


Figure 4. *Purification outcome plates. Negative group (N1) contained original strain with AN8455 ORF. Positive groups (P1–P3) contained transformants with Af-pyrG+. Among positive groups, P1 and P3 are used for subsequent PCR validation.*

4.3 Diagnostic PCR analysis

Diagnostic PCR was performed to determine whether samples N1, P1, and P3 were consistent with the original AN8455 locus or with the predicted Af-pyrG replacement structure. Based on the cassette design described in section 4.1, the two primer pair combinations are expected to obtain PCR products of 1124 bp and 1023 bp. On the diagnostic gel, 8 samples were added. S1 and S2 contained water controls, S3 and S4 contained N1 (original strain), S5 and S6 contained P1 (putative transformant 1), while S7 and S8 contained P3 (putative transformant 2) (Figure 5A). Before band-size estimation, a log-linear calibration was established from the DNA marker using migration distance and known fragment size according to Equation (1) (Figure 5B). The resulting model gave an R^2 of approximately 0.97, indicating that the calibration was sufficiently reliable for gel-based band-size prediction.

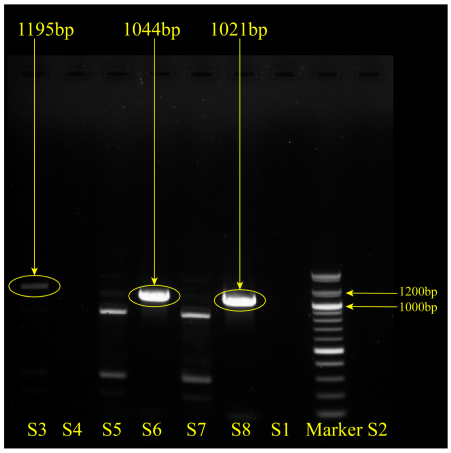
$$log_{10}(bp) = a + b * distance \quad (1)$$

To investigate our results (Figure 5, Table 1), the water controls (S1 and S2) showed no contamination and provided a clean background for interpreting the remaining lanes. The original-strain control N1 showed the expected native-locus pattern. In S3, a single band was detected at an estimated size of 1195 bp, close to the predicted 1124 bp product (+6.93%, acceptable). In S4, no visible band was observed. Together, these results are consistent with retention of the native AN8455 locus and absence of Af-pyrG insertion in the original strain. By contrast, both putative transformants showed patterns distinct from the original strain and consistent with cassette replacement. For P1, S5 produced two smaller non-target bands at approximately 779 bp and 239 bp, whereas S6 yielded a strong band at 1044 bp, close to the predicted 1023 bp transformant product. A closely similar pattern was observed for P3. In S7, two smaller bands were detected at approximately 788 bp and 231 bp, whereas S8 produced a strong band at 1021 bp, almost identical to the predicted 1023 bp product. The deviations of the transformant-specific bands in S6 and S8 from the predicted size were

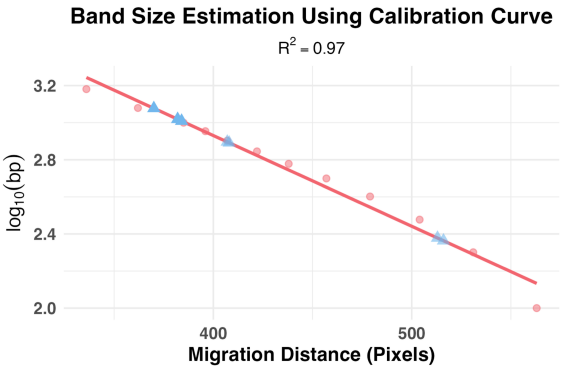
+2.05% and −0.20%, respectively. These small deviations, together with the absence of the expected native-locus band and the presence of the expected cassette-specific band, strongly support successful replacement of the endogenous AN8455 ORF in both P1 and P3. The off-target bands in S5 and S7 will be analyzed in section 4.4.

Band intensity differed across lanes, with the S3 band appearing noticeably weaker than the major bands in S6 and S8. However, intensity should not be directly compared across these reactions, because different reverse primers were used. Differences in primer structure, including GC content, secondary structure propensity, and related amplification properties, could alter PCR efficiency and thereby affect band brightness.

Taken together, the diagnostic PCR results support the conclusion that P1 and P3 are consistent with the predicted Af-pyrG replacement genotype, whereas N1 remains consistent with the original AN8455 locus. Overall, these data indicate that the endogenous AN8455 ORF was most likely successfully replaced by the Af-pyrG cassette in both putative transformants.



(A)



(B)

Figure 5. *Diagnostic PCR results.* ***(A)*** *The gel photo with annotated samples added in each lane. S3 (N1 + AN8455-F + AN8455-R) showed a band at 1195 bp (expected 1124 bp). S6 and S8 (P1/P3 + AN8455-F + Af-pyrG-R) showed bands at 1044 bp and 1021 bp, respectively (expected 1124 bp).* ***(B)*** *Calibration curve and band size estimation. Red points are marker bands used to fit the red regression line (R^2 = 0.97). Blue triangles are sample bands. Solid ones are desired products on S3/S6/S8, while transparent ones are off-target bands.*

Sample	*Expected Band Size (bp)*	*Estimated Band Size (bp)*
S1	*0*	*0*
S2	*0*	*0*
S3	*1124*	*1195*
S4	*0*	*0*
S5	*0*	*779 & 239*
S6	*1023*	*1044*
S7	*0*	*788 & 231*
S8	*1023*	*1021*

Table 1. *Expected band size in bp and estimated band size in bp for each sample.*

4.4 Band pattern analysis

In the gel image (Figure 5A), lanes S5 and S7 each showed two clear unexpected bands, and the corresponding band sizes were highly similar between the two samples (239/231 bp and 779/788 bp, respectively). The repeated appearance of two comparably sized bands in both putative transformants suggests that these products are unlikely to be random artifacts and may instead reflect some biological implications. Based on the analyses described below, these bands were provisionally interpreted as products generated by two non-specific annealing sites of AN8455-R on the transformant template. AN8455-R was inferred to bind non-specifically and yield two classes of minor PCR products with approximate mean sizes of 235 bp and 784 bp (mean of observed bands).

As the AN8455-F sequence remained present within the 5′ flank after cassette replacement, and no additional AN8455-F-like sequence could be identified within the Af-pyrG gene when up to five mismatches were allowed, the original AN8455-F was retained as the most plausible and effectively unique forward initiation site in the PCR system. The subsequent analysis therefore focused on potential non-specific binding sites of AN8455-R within transformant sequence. Using progressively relaxed mismatch thresholds, two candidate 20 bp sequences were identified when up to eight mismatches were permitted (12/20 matched bases; 60% identity): one starts at position 367 bp and one at 805 bp (Appendix IV). Their sequences (in reverse strand) could partially complementarily pair with AN8455-R directly, and were denoted as: AN8455-R-non-specific-short (AN8455-R_NS_S), with sequence *5'-ATCTTCAGCCTTATCAGCCA-3'*, and AN8455-R-non-specific-long (AN8455-R_NS_L), with sequence *5'-CTCTCCACCTCCAACTTGTA-3'*. When paired with AN8455-F, these candidate sites predicted amplicon sizes of 178 bp and 616 bp, respectively. These differed from the observed mean band sizes by 24.26% and 21.43%. Although not exact, these deviations remain within a plausible range.

Given that PCR outcome is influenced not only by perfect complementarity, but also by overall binding stability, terminal pairing stability, and cycling conditions (14-16). This provided the theoretical basis for further evaluation of the non-specific annealing hypothesis.

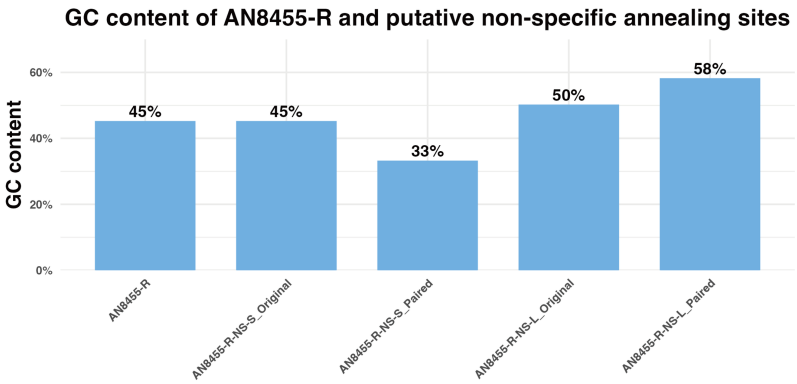
Our implication was then assessed from the properties of the two candidate non-specific annealing sites. Both candidate sites contained eight mismatches to AN8455-R, corresponding to 12 matched bases out of 20 (60% identity). Their GC contents were 45% for AN8455-R-NS-S and 50% for AN8455-R-NS-L, both equal to or higher than that of AN8455-R itself (45%), even after non-perfect pair with AN8455-R, GC contents for matched part were still 33% and 58%, respectively (Figure 6A). The relatively high GC proportion may be more stable and therefore increase the likelihood of non-specific binding. In addition, local alignment analysis (Figure 6B) showed that both candidate sites aligned well with AN8455-R with a larger proportion of continuous matches. More importantly, the alignment patterns were particularly well with the 5′ region of AN8455-R. Given that the candidate sites are reverse-complement sequences of the reverse primer, this corresponds to pairing between the 3′ end of the template and the 5′ region of the primer in the actual annealing event. Within the terminal six bases of the candidate sites, the complementary rate reached 83.33% in both cases, with relatively continuous matching. For a reverse primer, stability at the primer 3′ end is especially critical for productive extension, and this strong terminal compatibility further supports the possibility that these sites could support non-specific amplification.

Annealing temperature was then considered as an additional factor. In ApE, the fully complementary Tm values of AN8455-R, AN8455-R-NS-S, and AN8455-R-NS-L were predicted to be 51.7°C, 54.7°C, and 54.5°C, respectively. Although the actual interactions between AN8455-R and the candidate non-specific

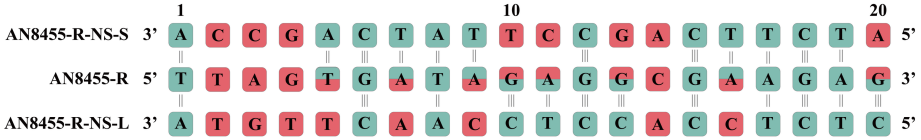
sites are mismatched rather than fully complementary, the intrinsically higher predicted Tm values of AN8455-R-NS-S and AN8455-R-NS-L suggest that annealing at 45°C could still permit stable non-specific binding. Under such a relatively low-temperature condition, these duplexes may fail to dissociate before extension and could therefore produce detectable PCR bands.

One remaining concern is why these putative non-specific sites lie within the 5′ flank region shared by both transformants and the original strain, whereas the original-strain sample did not show corresponding strong non-specific bands. A plausible explanation is that in the original strain, AN8455-R has access to its perfectly complementary target site, whose binding stability is much higher than that of the two non-specific sites. Once a substantial proportion of primer molecules is consumed by the fully matched target, the residual free primer concentration may become too low to support visible amplification from weaker non-specific sites.

Taken together, these analyses support the interpretation that the two unexpected bands observed in S5 and S7 were most likely generated by AN8455-R mediated non-specific annealing on two specific sites inside transformant template under a low annealing temperature, rather than random occurrences.



(A)



(B)

Figure 6. *Analyses for non-specific annealing sites.* ***(A)*** *GC contents comparison, including original AN8455-R diagnostic primer, AN8455-R-NS-S/L, in their original state and paired state with AN8455-R.* ***(B)*** *Specific alignment patterns between AN8455-R and AN8455-R-NS-S/L. Green denotes successful alignment, while red for failure. The alignment was especially good in 3' of AN8455-R (83.33% successful rate in the first 6 bp).*

5. Conclusion & Discussion

5.1 Conclusion of current reverse genetics workflow

Overall, the reverse-genetics workflow was successful through cassette design, transformation screening, purification-based candidate selection, genotype validation, and bioinformatics interpretation, although phenotype-level functional characterisation of AN8455 remains to be completed.

First, the knockout cassette design was successfully reconstructed in silico. Sequence annotation defined the genomic organization of the native AN8455 locus, the positions of the flanking arms and their associated primers, the endogenous ORF to be replaced by the Af-pyrG selectable marker through homologous recombination, and several diagnostic primers used for genotype validation. Second, the transformation stage produced limited but usable candidate material. Although Group 35 did not recover transformants, colonies obtained from parallel group material still demonstrated the overall outcome of the transformation experiment. Among the six colonies observed on the treatment plates, the colony from T2 showed morphology most consistent with the expected transformant and was therefore purified. Third, morphological comparison distinguished colonies with stable, Aspergillus-like growth patterns (N1, P1, and P3) from a colony with irregular, heterogeneous, and contamination-like features (P2) after purification. Fourth, diagnostic PCR provided the evidence that transformation had succeeded. The original strain control yielded a 1195 bp band, close to the predicted 1124 bp product (+6.93%), whereas the two putative transformants yielded 1044 bp and 1021 bp bands, consistent with the expected 1023 bp transformant product (+2.05% and −0.20%). Finally, bioinformatics analysis reviews information of AN8455, a conserved but still uncharacterised fungal gene. It has condition-dependent transcriptional regulation and a predicted intracellular, non-secreted, non-membrane protein product.

Overall, this study successfully completed the design-to-genotype portion of the reverse-genetics workflow, whereas the final genotype-to-phenotype step remains to be addressed in future work.

5.2 Review of unexpected observations

Two unexpected observations emerged during the workflow and required further interpretation: the failure of the Group 35 transformation plates to yield usable colonies, and the appearance of additional non-target bands in the AN8455-F + AN8455-R PCR reactions of the transformant samples.

For the first observation, successful transformation in parallel groups suggests that the failure was unlikely to arise from knockout-cassette design or reagent preparation. It was therefore more plausibly attributed to a combination of manipulation error and intrinsic fragility within the protoplast-mediated transformation process. One major source of failure may have occurred during protoplast preparation. During cell wall digestion, excessive or prolonged enzymatic treatment can damage not only the wall but also cell membrane integrity, thereby increasing osmotic fragility and reducing protoplast viability. Such cells may lyse during subsequent PEG exposure and fail to regenerate a cell wall after plating, resulting in no colony recovery despite exposure to DNA (17). Furthermore, even correctly prepared protoplasts remain highly fragile, and minor osmotic imbalance during handling, can rapidly trigger lysis (18). In addition, successful DNA uptake does not guarantee colony formation, because transformed protoplasts must still survive the recovery phase and regenerate a functional wall before cell division begins. Stress at this stage, including suboptimal top-layer temperature, delayed mixing, or uneven plating, could further reduce survival (19). Another likely reason for failure is reagent-dependent transformation efficiency, particularly the use of PEG, which is central to DNA uptake. As transformation depends on transient membrane permeability induced by PEG, non-optimal PEG viscosity, mixing, or incubation timing can sharply reduce DNA entry and therefore lower transformation frequency (20).

Furthermore, these factors are unlikely to act independently. Errors at multiple stages may combine multiplicatively and reduce overall efficiency to zero, which is consistent with the complete absence of usable colonies on the Group 35 transformation plates.

The second observation, namely the appearance of two off-target bands in both transformant samples, is more consistent with reproducible non-specific amplification than with random PCR noise. The most plausible explanation is that the transformant sequence contains two candidate non-specific annealing sites for AN8455-R, which became amplifiable under the suboptimal PCR annealing temperature. As shown in section 4.4, the predicted products generated from these two candidate sites showed deviations of 24.26% and 21.43% from the mean band sizes observed in S5 and S7, providing initial support for this interpretation. More importantly, the candidate sites were further evaluated at the sequence level, including GC content, overall alignment with AN8455R, predicted Tm, and compatibility with the PCR conditions. It is noteworthy that well terminal alignment interpreted this implication a lot, because productive primer extension depends strongly on stability at the 3′ end. In both candidate sites, complementarity across the terminal six bases reached 83.33%, providing strong theoretical support for their ability to function as non-specific primer-binding sites.

5.3 Limitations and future works

Although the present study successfully completed the design-to-genotype portion of the reverse-genetics workflow, several limitations remain and should be addressed in future work.

The most important limitation is that phenotype-level functional analysis has not yet been completed. In reverse genetics, functional inference ultimately depends on comparison of the original strain with the mutant strain at the phenotypic level. Therefore, current priority is systematic phenotypic screening of N1 and the validated transformants (P1/P3), together with their further reproduced strains, in order to extend the study to a more complete genotype-to-phenotype interpretation. Relevant assays should include colony and hyphal morphology, growth kinetics, survival under stress conditions such as osmotic, oxidative, and cell wall stress, and secondary metabolite profiles, so that the physiological role of AN8455 can be assessed more directly.

Secondly, a complementary strategy would be to combine reverse and forward genetic approaches. For example, AN8455 could be reintroduced into an AN8455-knockout strain, followed by phenotypic testing to determine whether the original phenotype can be restored. Such a complementation design would provide stronger evidence that any observed phenotype is specifically attributable to loss of AN8455 rather than to background effects introduced during transformation.

Computational approaches could also be incorporated. One possible direction is to model primer-template duplex formation under PCR-like temperature conditions, for example, by simulating thermodynamic behavior, estimating binding free energy (ΔG), and monitoring hydrogen-bonding patterns, in order to evaluate the plausibility of the proposed off-target products. In parallel, the predicted AN8455 protein structure could be analyzed computationally, including structural simulation under different conditions or docking-based prediction of interaction modes, to generate hypotheses about its molecular function.

The two unexpected observations discussed in section 5.2 also warrant further study. To improve transformation efficiency, alternative approaches could be considered, including CRISPR/Cas9-based gene editing, which may offer higher precision and reduce dependence on protoplast stability (21). Further validation of the off-target bands could be pursued in two ways. The most direct and reliable approach would be gel extraction and sequencing of

the two unexpected bands in S5 and S7, followed by alignment of the recovered sequences to the original target region and cassette sequence. However, given that this approach may be costly, a practical alternative would be gradient-PCR optimization of annealing temperature. Increasing the annealing temperature stepwise from 45°C to values exceeding the predicted tolerance of the putative non-specific sites would provide a straightforward test of whether these bands are indeed driven by low-temperature non-specific annealing.

6. Reference

1. Benfey, P.N. and Mitchell-Olds, T. (2008) From genotype to phenotype: systems biology meets natural variation. Science, 320, 495-497.
2. Sahu, P.K., Sao, R., Mondal, S., Vishwakarma, G., Gupta, S.K., Kumar, V., Singh, S., Sharma, D. and Das, B.K. (2020) Next Generation Sequencing Based Forward Genetic Approaches for Identification and Mapping of Causal Mutations in Crop Plants: A Comprehensive Review. Plants (Basel), 9.
3. Bhadauria, V., Banniza, S., Wei, Y. and Peng, Y.L. (2009) Reverse genetics for functional genomics of phytopathogenic fungi and oomycetes. Comp Funct Genomics, 2009, 380719.
4. McCluskey, K., Wiest, A. and Plamann, M. (2010) The Fungal Genetics Stock Center: a repository for 50 years of fungal genetics research. J Biosci, 35, 119-126.
5. Szewczyk, E., Nayak, T., Oakley, C.E., Edgerton, H., Xiong, Y., Taheri-Talesh, N., Osmani, S.A. and Oakley, B.R. (2006) Fusion PCR and gene targeting in Aspergillus nidulans. Nature Protocols, 1, 3111-3120.
6. Golduran, G.H. and Morris, N.R. (1995) In Adolph, K. W. (ed.), Methods in Molecular Genetics. Academic Press, Vol. 6, pp. 48-65.
7. Galagan, J.E., Calvo, S.E., Cuomo, C., Ma, L.-J., Wortman, J.R., Batzoglou, S., Lee, S.-I., Baştürkmen, M., Spevak, C.C., Clutterbuck, J. et al. (2005) Sequencing of Aspergillus nidulans and comparative analysis with A. fumigatus and A. oryzae. Nature, 438, 1105-1115.
8. Dyer, P.S. and O'Gorman, C.M. (2012) Sexual development and cryptic sexuality in fungi: insights from Aspergillus species. FEMS Microbiology Reviews, 36, 165-192.
9. Hoffmann, B., Eckert, S.E., Krappmann, S. and Braus, G.H. (2001) Sexual diploids of Aspergillus nidulans do not form by random fusion of nuclei in the heterokaryon. Genetics, 157, 141-147.
10. Ninomiya, Y., Suzuki, K., Ishii, C. and Inoue, H. (2004) Highly efficient gene replacements in Neurospora strains deficient for nonhomologous end-joining. Proc Natl Acad Sci U S A, 101, 12248-12253.
11. Stajich, J.E., Harris, T., Brunk, B.P., Brestelli, J., Fischer, S., Harb, O.S., Kissinger, J.C., Li, W., Nayak, V., Pinney, D.F. et al. (2012) FungiDB: an integrated functional genomics database for fungi. Nucleic Acids Res, 40, D675-681.
12. Weidner, G., d'Enfert, C., Koch, A., Mol, P.C. and Brakhage, A.A. (1998) Development of a homologous transformation system for the human pathogenic fungus Aspergillus fumigatus based on the pyrG gene encoding orotidine 5'-monophosphate decarboxylase. Curr Genet, 33, 378-385.
13. Nayak, T., Szewczyk, E., Oakley, C.E., Osmani, A., Ukil, L., Murray, S.L., Hynes, M.J., Osmani, S.A. and Oakley, B.R. (2006) A versatile and efficient gene-targeting system for Aspergillus nidulans. Genetics, 172, 1557-1566.
14. Kwok, S., Kellogg, D.E., McKinney, N., Spasic, D., Goda, L., Levenson, C. and Sninsky, J.J. (1990) Effects of primer-template mismatches on the polymerase chain reaction: Human immunodeficiency virus type 1 model studies. Nucleic Acids Research, 18, 999-1005.
15. SantaLucia, J. (1998) A unified view of polymer, dumbbell, and oligonucleotide DNA nearest-neighbor thermodynamics. Proceedings of the National Academy of Sciences, 95, 1460-1465.

16. Rubin, E. and Levy, A.A. (1996) A mathematical model and a computerized simulation of PCR using complex templates. Nucleic Acids Res, 24, 3538-3545.
17. Davis, B. (2020), pp. 45-71.
18. Yelton, M.M., Hamer, J.E. and Timberlake, W.E. (1984) Transformation of Aspergillus nidulans by using a trpC plasmid. Proceedings of the National Academy of Sciences, 81, 1470-1474.
19. Fincham, J.R. (1989) Transformation in fungi. Microbiological Reviews, 53, 148-170.
20. Ballance, D.J. (1991) Transformation systems for filamentous fungi and an overview of fungal gene structure. Mycology series., 8, 1-29.
21. Nødvig, C., Hoof, J., Kogle, M. and Mortensen, U. (2015) A CRISPR-Cas9 system for genetic engineering of filamentous fungi. PloS one, 10, e0133085.

7. Appendix

Appendix I. Bioinformatics Data Analysis Sheet

Data Analysis Sheet – Reverse Genetics Experiment for AN8455

Author: Yuewei Xu **Data:** 2026-04-03

Gene Information

- **Gene Name:** Uncharacterised
- **Systematic Name:** AN8455, ANID_08455
- **Gene Type:** Protein Coding Gene
- **Description:** AN8455 (ANID_08455) is a protein-coding gene in *Aspergillus nidulans* FGSC A4 strain located on the negative strand of chromosome V (541006–543575). Orthologs are present in several Aspergillus species, indicating conservation within this lineage. No specific molecular function has yet been assigned in the annotation. However, transcriptomic data suggest condition-dependent regulation, including differential expression across nitrogen sources, developmental stages, and selected regulatory mutant backgrounds. The gene contains 8 exons and 1 annotated transcript (AN8455-T), encoding a protein of 653 amino acids, which is predicted to be intracellular, non-secreted, and non-transmembrane and most likely localized to the cytoplasm or nucleus. Overall, AN8455 appears to be a conserved but uncharacterised fungal gene whose role requires experimental validation by reverse genetics.
- **Reference:** For *A. nidulans*, no gene-specific literature is available.
- **Genome Location:** ChrV_A_nidulans_FGSC_A4, 541006 - 543575 in Reverse Strand
- **Annotated DNA Sequence:** Annotated sequence can be retrieved in Appendix II, where 1200 bp upstream and downstream the gene were included.
- **Orthologs:**
    - A. fumigatus Af293: Afu5g10940, Afu5g10960
    - Neosartorya fischeri NRRL 181: NFIA_076300
    - Aspergillus wentii: Aspwe1_0041244
    - Aspergillus versicolor: Aspve1_0042580

Transcriptomics Information

- **Transcript:** 1 transcript (AN8455-T): an mRNA formed by 8 exons with a total length of 1980 nucleotides.
- **Transcriptomics (RNA-seq) Analysis:**
    - AN8455 is differentially expressed across conditions:
    - Ammonium-rich conditions: significantly upregulated – nitrogen-linked regulatory.
    - Deletion of veA or mtfA genes: upregulated – pathways linked to regulatory.
    - Complete media conditions: significantly upregulated – sensitivity to nutrition supply.
    - Development phase: upregulated in hyphae relative to conidia – related to fungi growth.
    - Crystalline cellulose exposure: moderate downregulation – independent of carbon metabolism.
- **Functional Analysis (GO Annotation Sourced from AspGD):**
    - GO:0008150: biological_process
    - GO:0005575: celluar_component
    - GO:0003674: molecular_function
    - Only three root terms of GO annotation are listed.

Protein Information

- **Predicted Protein Sequence:**
  MSDQYSLSHLLEQQHLADCTKYERQWRPSDDNSPPPNEPVGPGTMRDVREEQFYLGGKNTVQPWDDSFP SPLMPAIYDPYPDYNSGVWSDTWKGTFRTCEGPRGGTLDRQNVEDMVLTYPGVQGGFPYPKYGSYEALGL DGYTCTTRASRLAAYGYSESGLNSTADPIVWDNVNWGTLQSECLQRNINRYHEPKRTPRLSILPLNPIRTEAPA NDDEGISSPATKKRTAVVLRAWHDIVWTENLKEHVRSLIMELSLHSGAEYEVFLLTHVKDNDLLLCGVDEAKI RALKKKFIPREFWDITVLFDERTLQSWYPKVDEHRPVYQYAQPLQIFSTVFPDFHYYWQLEMDARFTGHTY HFLERASEFAKQQSRKYLWERNAYFYIPGAHGTWEEFMHTVDKSLLKRESNTVWGPAPPHSWIQPVGPEPPI SSPQDDDYEWGVGEEADLITFLPIFDPSDTTWVFWNLLWNLPIETTTRRTSPVMIGRFSHRLLQVMHNAQIE QGVALVSEMMPSSFALWHGLKAVHAPHPIYADGKWMPKELNNIVNKVAGGPEKINGGKDSFWNWDHKLD HIVYRMSYMFTGQPAEDFYRRWLGYKPNPAQYTDGSRDPQGRNWFDGGHLREDIYGPLCFPSMLLHPIKN TELDKGPDMAVPV
- **Protein Features and Properties:**
  - Length: 653 Amino Acids
  - Isoelectric Point: 5.75
  - Molecular Weight: 75391 Da
  - no Signal Peptide (SignalP)
  - no transmembrane helix (TMHMM)
- **Detailed Test Results:**
  - **Pfam Domain:** One major conserved domain (PF11885 – DUF3405 Protein of unknown function DUF3405) located at amino acids 143–640 with highly significant matches (E-value = 0) and corresponds to InterPro entry IPR021822.
  - **GPI-anchor prediction (big-PI Predictor in Fungi):** no potential GPI-modification site – no glycosylphosphatidylinositol (GPI) anchored surface or cell-wall-associated role for AN8455.
  - **WoLF PSORT:** most likely cytoplasmic, although a possible nuclear localisation was also suggested (queryProtein WoLFPSORT prediction cyto: 10.5, nucl: 9, cyto_pero: 8, pero: 4.5, mito: 2, vacu: 1).
  - **MitoProt:** no predictable mitochondrial targeting sequence (PROBABILITY of export to mitochondria: 0.0646) –unlikely to be imported into mitochondria (consistent with WoLF PSORT score for mito).
  - The protein encoded by AN8455 is predicted to be intracellular, non-secreted, and non-transmembrane.
- **Predicted Structure (AlphaFold 3):**
  - Predicted structure can be seen in Figure 7A.
  - Overall pTM score is 0.93 – high confidence for the structural accuracy

Af-pyrG Information

- **Annotated DNA Sequence:** Annotated sequence can be retrieved in Appendix III
- **Predicted Protein Sequence:** 278 Amino Acids -
  MSSKSQLTYGARASKHPNPLAKRLFEIAEAKKTNVTVSADVTTTRELLDLADRLGPYIAVIKTHIDILTDFSVDT INGLNVLAQKHNFLIFEDRKFIDIGNTVQKQYHGGALRISEWAHIINCSVLPGEGIVEALAQTASAQDFPYGP ERGLLVLAEMTSKGSLATGEYTKASVDYARKYKNFVMGFVSTRALTEVQSDVSSASEDEDFVVFTTGVNLSSK GDKLGQQYQTPASAIGRGADFIIAGRGIYAAPDPVEAAQRYQKEGWEAYMARVCGKS
- **Predicted Structure (AlphaFold 3):**
  - Predicted structure can be seen in Figure 7B.

- Overall pTM score is 0.95 – high confidence for the structural accuracy

- **Expected PCR Product Size:** 1124 bp

Experiment Related Information

- **Primers for Deletion Cassette:**
  - Blue sequences are consensus sites.
  - Black sequences are AN8455-specific deletion cassette primers.
  - 5' Forward: *5'-GTAACGCCAGGGTTTTCCCAGTCACGACGGGGTGAAGGGATACAGTTAC-3'*
  - 5' Reverse: *5'-ATCCACTTAACGTTACTGAAATCGATCAGACATGGAGGACAC-3'*
  - 3' Forward: *5'-CTCCTTCAATATCATCTTCTGTCCTGATATGGCAGTCCCAGT-3'*
  - 3' Reverse: *5'-GCGGATAACAATTTCACACAGGAAACAGCAGGAGTGTAAGGCTGAAGAG-3'*
- **Primers for Diagnosis:**
  - ANID_08455 Forward: *5'-TAAGGCGAGAATAAAGAAA-3'*
  - ANID_08455 Reverse: *5'-TTAGTGATAGAGGCGAAGAG-3'*
- **Expected PCR Product Size:** 1124 bp

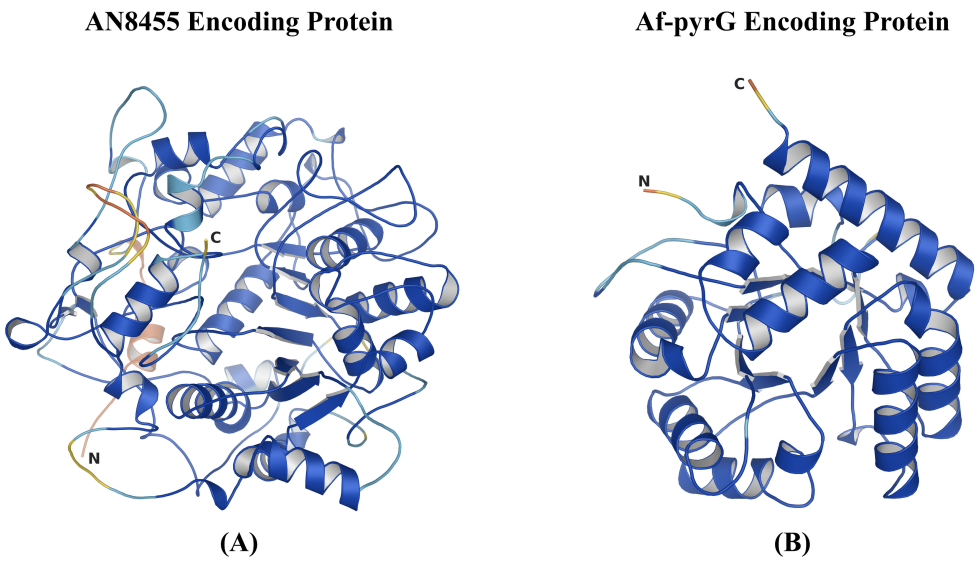
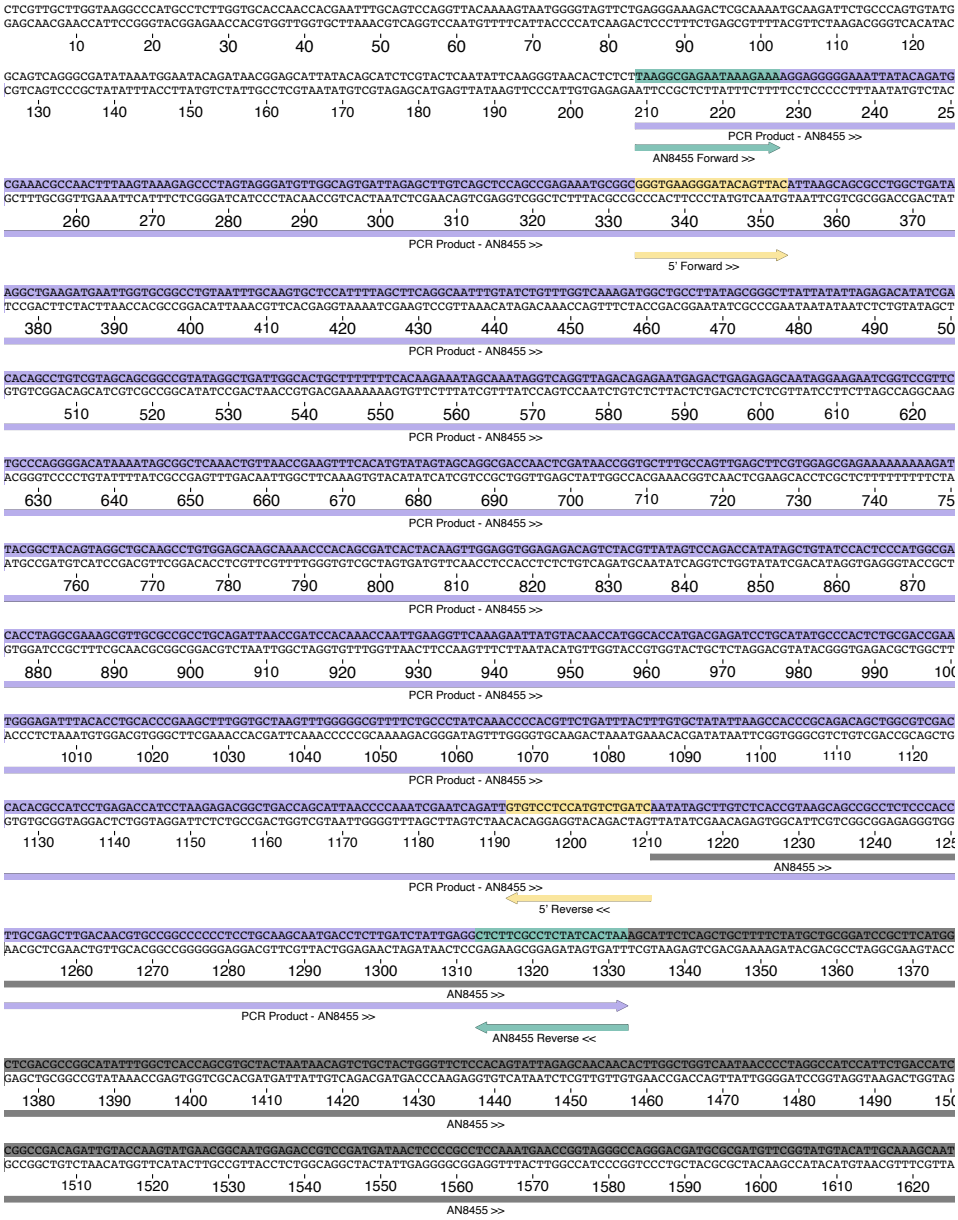
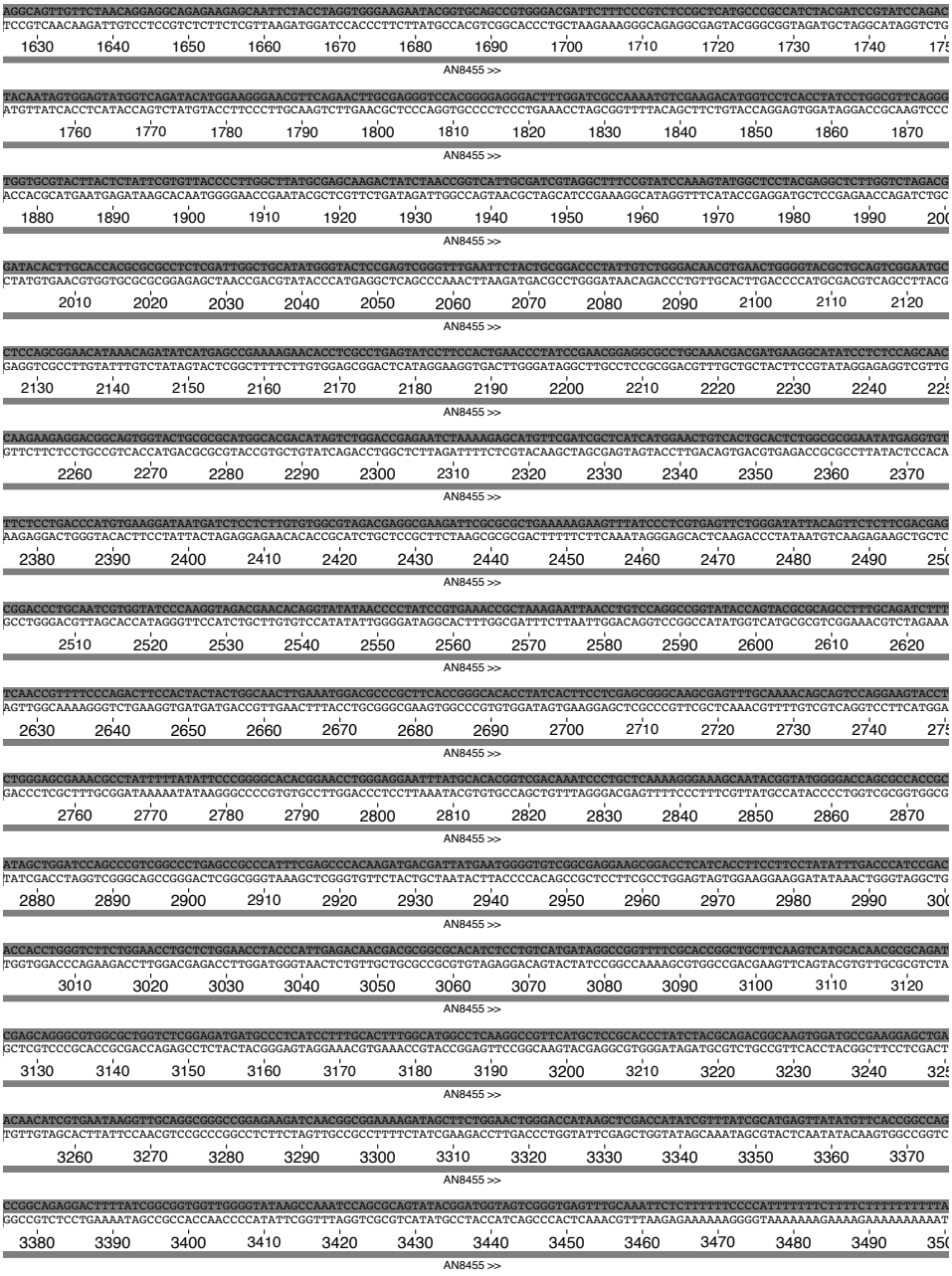
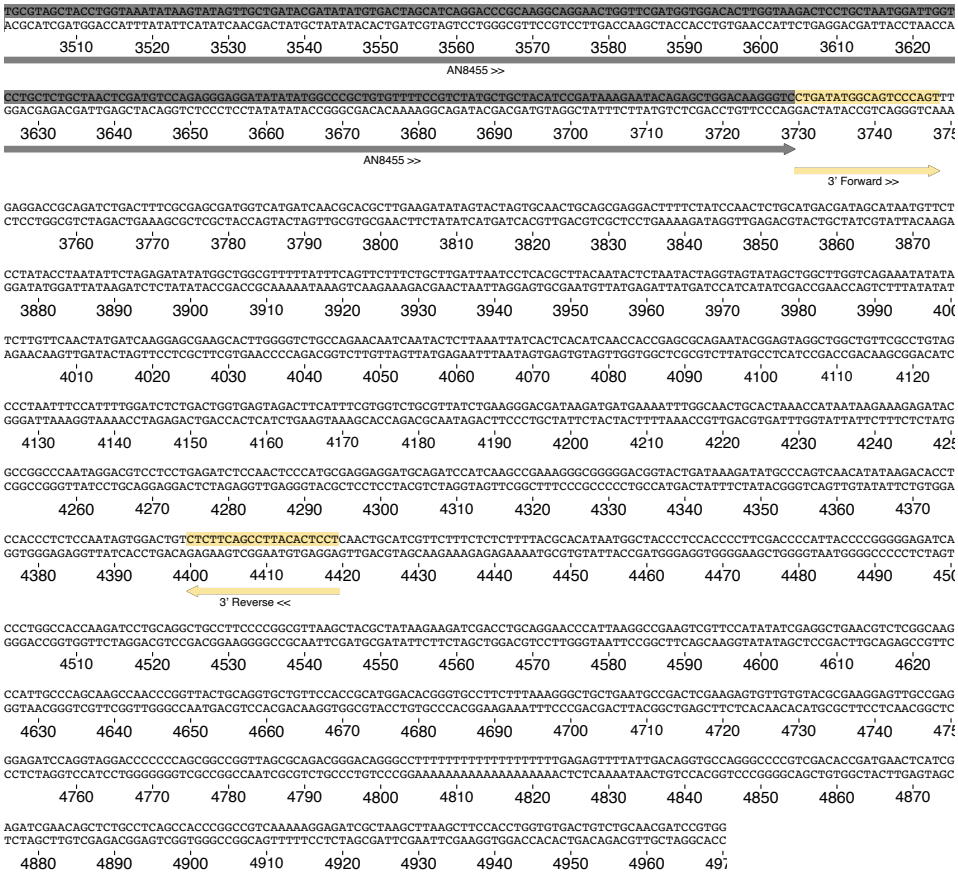


Figure 7. *AlphaFold3 predicted protein structures. Colored by confidence level of each residue: dark blue for very high confidence, light blue for high, yellow for low, and orange for very low.* ***(A)*** *Protein encoded by AN8455 (Overall pTM score: 0.93).* ***(B)*** *Protein encoded by Af-pyrG+ (Overall pTM score: 0.95).*

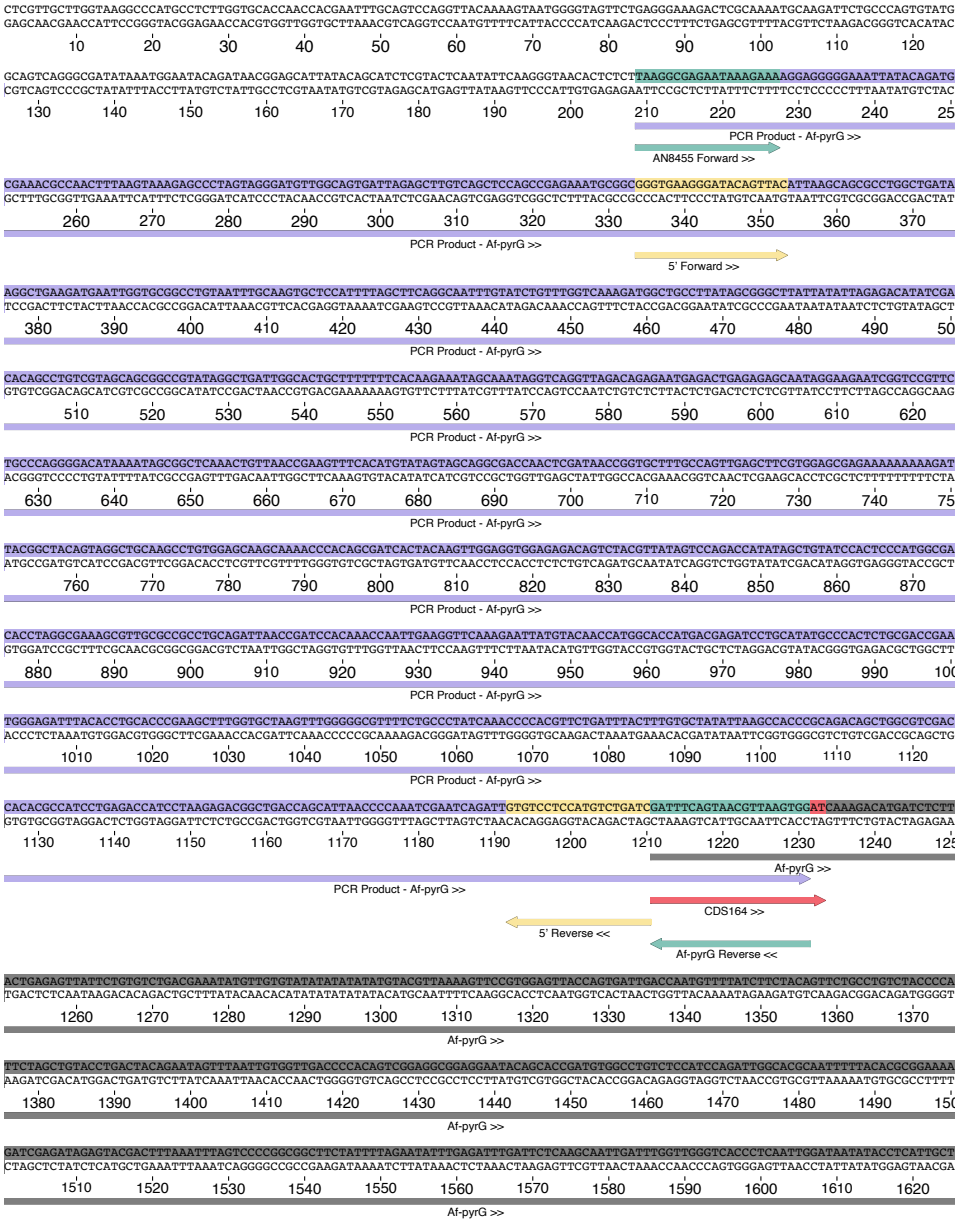
Appendix II. Annotated DNA Sequence for AN8455 locus

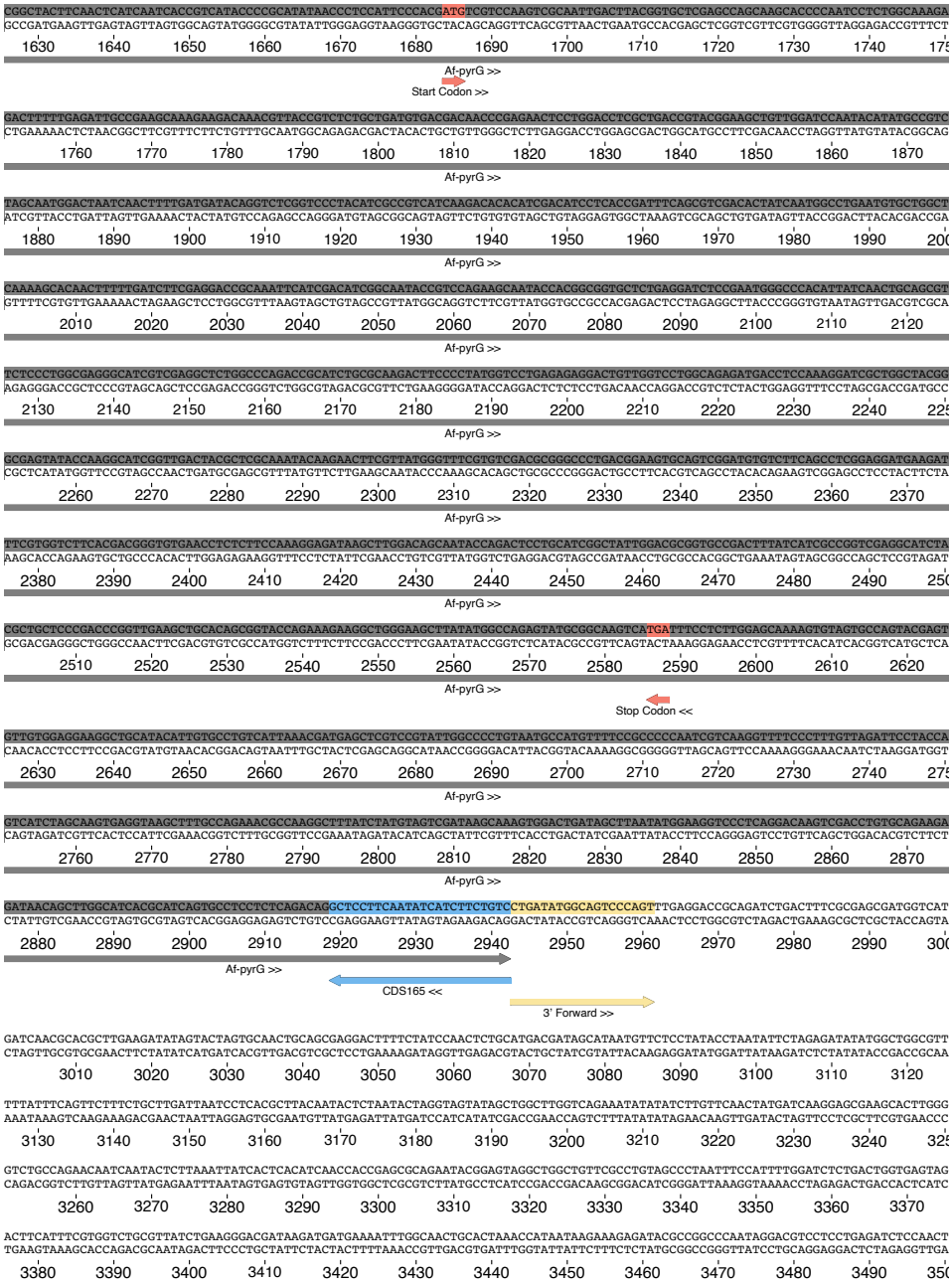


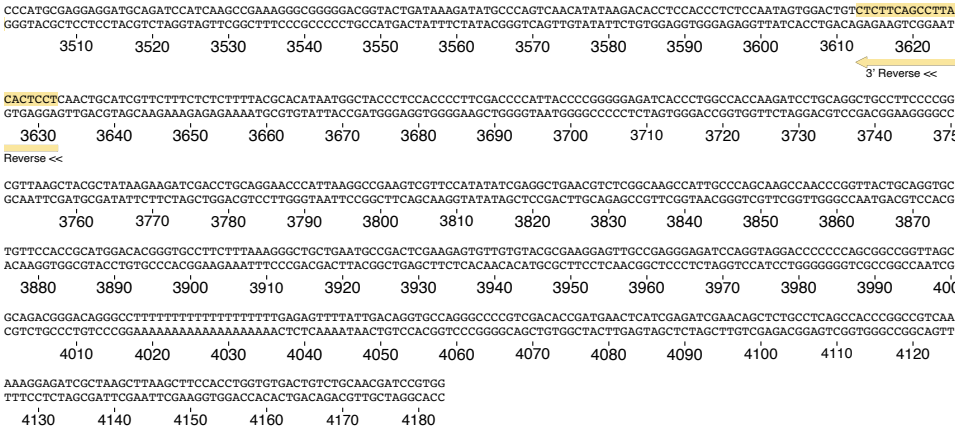

AGGCAGTTGTTCTAACAGGAGGCAGAGAAGAGCAATTCTACCTAGGTGGGAAGAATACGGTGCAGCCGTGGGACGATTCTTTCCCGTCTCCGCTCATGCCCGCCATCTACGATCCGTATCCAGAC
TCCGTCAACAAGATTGTCCTCCGTCTCTTCTCGTTAAGATGGATCCACCCTTCTTATGCCACGTCGGCACCCTGCTAAGAAAGGGCAGAGGCGAGTACGGGCGGTAGATGCTAGGCATAGGTCTG
1630 1640 1650 1660 1670 1680 1690 1700 1710 1720 1730 1740 175
AN8455 >>
TACAATAGTGGAGTATGGTCAGATACATGGAAGGGAACGTTCAGAACTTGCGAGGGTCCACGGGGAGGGACTTTGGATCGCCAAAATGTCGAAGACATGGTCCTCACCTATCCTGGCGTTCAGGG
ATGTTATCACCTCATACCAGTCTATGTACCTTCCCTTGCAAGTCTTGAACGCTCCCAGGTGCCCCTCCCTGAAACCTAGCGGTTTTACAGCTTCTGTACCAGGAGTGGATAGGACCGCAAGTCCC
1760 1770 1780 1790 1800 1810 1820 1830 1840 1850 1860 1870
AN8455 >>
TGGTGGCTACTTACTCTATTCCTGTTACCCCTTGGCTTATGCGAGCAAGACTATCTAACCGGTCATTGCGATCGTAGGCTTTCCGTATCCAAAGTATGGCTCCTACGAGGCTCTTGGTCTAGACG
ACCACGCATGAATGAGATAAGCACAATGGGGAACCGAATACGCTCGTTCTGATAGATTGGCCAGTAACGCTAGCATCCGAAAGGCATAGGTTTCATACCGAGGATGCTCCGAGAACCAGATCTGC
1880 1890 1900 1910 1920 1930 1940 1950 1960 1970 1980 1990 200
AN8455 >>
GATACACTTGCACCACGCGCGCCTCTCGATTGGCTGCATATGGGTACTCCGAGTCGGGTTTGAATTCTACTGCGGACCCTATTGTCTGGGACAACGTGAACTGGGGTACGCTGCAGTCGGAATGC
CTATGTGAACGTGGTGCGCGCGGAGAGCTAACCGACGTATACCCATGAGGCTCAGCCCAAACTTAAGATGACGCCTGGGATAACAGACCCTGTTGCACTTGACCCCATGCGACGTCAGCCTTACG
2010 2020 2030 2040 2050 2060 2070 2080 2090 2100 2110 2120
AN8455 >>
CTCCAGCGGAACATAAACAGATATCATGAGCCGAAAAGAACACCTCGCCTGAGTATCCTTCCACTGAACCCTATCCGAACGGAGGCGCCTGCAAACGACGATGAAGGCATATCCTCTCCAGCAAC
GAGGTCGCCTTGTATTTGTCTATAGTACTCGGCTTTTCTTGTGGAGCGGACTCATAGGAAGGTGACTTGGGATAGGCTTGCCTCCGCGGACGTTTGCTGCTACTTCCGTATAGGAGAGGTCGTTG
2130 2140 2150 2160 2170 2180 2190 2200 2210 2220 2230 2240 225
AN8455 >>
CAAGAAGAGGACGGCAGTGGTACTGCGCGCATGGCACGACATAGTCTGGACCGAGAATCTAAAAGAGCATGTTCGATCGCTCATCATGGAACTGTCACTGCACTCTGGCGCGGAATATGAGGTGT
GTTCTTCTCCTGCCGTCACCATGACGCGCGTACCGTGCTGTATCAGACCTGGCTCTTAGATTTTCTCGTACAAGCTAGCGAGTAGTACCTTGACAGTGACGTGAGACCGCGCCTTATACTCCACA
2260 2270 2280 2290 2300 2310 2320 2330 2340 2350 2360 2370
AN8455 >>
TTCTCCTGACCCATGTGAAGGATAATGATCTCCTCTTGTGTGGCGTAGACGAGGCGAAGATTCGCGCGCTGAAAAAGAAGTTTATCCCTCGTGAGTTCTGGGATATTACAGTTCTCTTCGACGAG
AAGAGGACTGGGTACACTTCCTATTACTAGAGGAGAACACACCGCATCTGCTCCGCTTCTAAGCGCGCGACTTTTTCTTCAAATAGGGAGCACTCAAGACCCTATAATGTCAAGAGAAGCTGCTC
2380 2390 2400 2410 2420 2430 2440 2450 2460 2470 2480 2490 250
AN8455 >>
CGGACCCTGCAATCGTGGTATCCCAAGGTAGACGAACACAGGTATATAACCCCTATCCGTGAAACCGCTAAAGAATTAACCTGTCCAGGCCGGTATACCAGTACGCGCAGCCTTTGCAGATCTTT
GCCTGGGACGTTAGCACCATAGGGTTCCATCTGCTTGTGTCCATATATTGGGGATAGGCACTTTGGCGATTTCTTAATTGGACAGGTCCGGCCATATGGTCATGCGCGTCGGAAACGTCTAGAAA
2510 2520 2530 2540 2550 2560 2570 2580 2590 2600 2610 2620
AN8455 >>
TCAACCGTTTTCCCAGACTTCCACTACTACTGGCAACTTGAAATGGACGCCCGCTTCACCGGGCACACCTATCACTTCCTCGAGCGGGCAAGCGAGTTTGCAAAACAGCAGTCCAGGAAGTACCT
AGTTGGCAAAAGGGTCTGAAGGTGATGATGACCGTTGAACTTTACCTGCGGGCGAAGTGGCCCGTGTGGATAGTGAAGGAGCTCGCCCGTTCGCTCAAACGTTTTGTCGTCAGGTCCTTCATGGA
2630 2640 2650 2660 2670 2680 2690 2700 2710 2720 2730 2740 275
AN8455 >>
CTGGGAGCGAAACGCCTATTTTTATATTCCCGGGGCACACGGAACCTGGGAGGAATTTATGCACACGGTCGACAAATCCCTGCTCAAAAGGGAAAGCAATACGGTATGGGGACCAGCGCCACCGC
GACCCTCGCTTTGCGGATAAAAATATAAGGGCCCCGTGTGCCTTGGACCCTCCTTAAATACGTGTGCCAGCTGTTTAGGGACGAGTTTTCCCTTTCGTTATGCCATACCCCTGGTCGCGGTGGCG
2760 2770 2780 2790 2800 2810 2820 2830 2840 2850 2860 2870
AN8455 >>
ATAGCTGGATCCAGCCCGTCGCCCCTGAGCCGCCCATTTCGAGCCCACAAGATGACGATTATGAATGGGGTGTCGGCGAGGAAGCGGACCTCATCACCTTCCTTCCTATATTTGACCCATCCGAC
TATCGACCTAGGTCGGGCAGCCGGGACTCGGCGGGTAAAGCTCGGGTGTTCTACTGCTAATACTTACCCCACAGCCGCTCCTTCGCCTGGAGTAGTGGAAGGAAGGATATAAACTGGGTAGGCTG
2880 2890 2900 2910 2920 2930 2940 2950 2960 2970 2980 2990 300
AN8455 >>
ACCACCTGGGTCTTCTGGAACCTGCTCTGGAACCTACCCATTGAGACAACGACGCGGCGCACATCTCCTGTCATGATAGGCCGGTTTTCGGCACCGGCTGCTTCAAGTCATGCACAACGCGCAGAT
TGGTGGACCCAGAAGACCTTGGACGAGACCTTGGATGGGTAACTCTGTTGCTGCGCCGCGTGTAGAGGACAGTACTATCCGGCCAAAAGCGTGGCCGACGAAGTTCAGTACGTGTTGCGCGTCTA
3010 3020 3030 3040 3050 3060 3070 3080 3090 3100 3110 3120
AN8455 >>
CGAGCAGGGCGTGGCGCTGGTCTCGGAGATGATGCCCTCATCCTTTGCACTTTGGCATGGCCTCAAGGCCGTTCATGCTCCGCACCCTATCTACGCAGACGGCAAGTGGATGCCGAAGGAGCTGA
GCTCGTCCCGCACCGCGACCAGAGCCTCTACTACGGGAGTAGGAAACGTGAAACCGTACCGGAGTTCCGGCAAGTACGAGGCGTGGGATAGATGCGTCTGCCGTTCACCTACGGCTTCCTCGACT
3130 3140 3150 3160 3170 3180 3190 3200 3210 3220 3230 3240 325
AN8455 >>
ACAACATCGTGAATAAGGTTGCAGGCGGGCCGGAGAAGATCAACGGCGGAAAAGATAGCTTCTGGAACTGGGACCATAAGCTCGACCATATCGTTTATCGCATGAGTTATATGTTCACCGGCCAG
TGTTGTAGCACTTATTCCAACGTCCGCCCGGCCTCTTCTAGTTGCCGCCTTTTCTATCGAAGACCTTGACCCTGGTATTCGAGCTGGTATAGCAAATAGCGTACTCAATATACAAGTGGCCGGTC
3260 3270 3280 3290 3300 3310 3320 3330 3340 3350 3360 3370
AN8455 >>
CCGGCAGAGGACTTTTATCGGCGGTGGTTGGGGTATAAGCCAAATCCAGCGCAGTATACGGATGGTAGTCGGGTGAGTTTGCAAATTCTCTTTTTTCCCCATTTTTTTCTTTTCTTTTTTTTTTA
GGCCGTCTCCTGAAAATAGCCGCCACCAACCCCATATTCGGTTTAGGTCGCGTCATATGCCTACCATCAGCCCACTCAAACGTTTAAGAGAAAAAAGGGGTAAAAAAAGAAAAGAAAAAAAAAT
3380 3390 3400 3410 3420 3430 3440 3450 3460 3470 3480 3490 350
AN8455 >>


AN8455 >>
AN8455 >>
3' Forward >>
3' Reverse <<

Appendix III. Annotated DNA Sequence for Af-pyrG+ Cassette




Af-pyrG >>
Start Codon >>
Af-pyrG >>
Af-pyrG >>
Af-pyrG >>
Af-pyrG >>
Af-pyrG >>
Stop Codon <<
Af-pyrG >>
Af-pyrG >>
Af-pyrG >>
CDS165 <<
3' Forward >>


CCCATGCGAGGAGGATGCAGATCCATCAAGCCGAAAGGGCGGGGGACGGTACTGATAAAGATATGCCCAGTCAACATATAAGACACCTCCACCCTCTCCAATAGTGGACTGTCTCTTCAGCCTTA
GGGTACGCTCCTCCTACGTCTAGGTAGTTCGGCTTTCCCGCCCCCTGCCATGACTATTTCTATACGGGTCAGTTGTATATTCTGTGGAGGTGGGAGAGGTTATCACCTGACAGAGAAGTCGGAAT
3510 3520 3530 3540 3550 3560 3570 3580 3590 3600 3610 3620
3' Reverse <<
CACTCCTCAACTGCATCGTTCTTTCTCTCTTTTACGCACATAATGGCTACCCTCCACCCCTTCGACCCCATTACCCCGGGGGAGATCACCCTGGCCACCAAGATCCTGCAGGCTGCCTTCCCCGG
GTGAGGAGTTGACGTAGCAAGAAAGAGAGAAAATGCGTGTATTACCGATGGGAGGTGGGGAAGCTGGGGTAATGGGGCCCCCTCTAGTGGGACCGGTGGTTCTAGGACGTCCGACGGAAGGGGCC
3630 3640 3650 3660 3670 3680 3690 3700 3710 3720 3730 3740 375
Reverse <<
CGTTAAGCTACGCTATAAGAAGATCGACCTGCAGGAACCCATTAAGGCCGAAGTCGTTCCATATATCGAGGCTGAACGTCTCGGCAAGCCATTGCCCAGCAAGCCAACCCGGTTACTGCAGGTGC
GCAATTCGATGCGATATTCTTCTAGCTGGACGTCCTTGGGTAATTCCGGCTTCAGCAAGGTATATAGCTCCGACTTGCAGAGCCGTTCGGTAACGGGTCGTTCGGTTGGGCCAATGACGTCCACG
3760 3770 3780 3790 3800 3810 3820 3830 3840 3850 3860 3870
TGTTCCACCGCATGGACACGGGTGCCTTCTTTAAAGGGCTGCTGAATGCCGACTCGAAGAGTGTTGTGTACGCGAAGGAGTTGCCGAGGGAGATCCAGGTAGGACCCCCCCAGCGGCCGGTTAGC
ACAAGGTGGCGTACCTGTGCCCACGGAAGAAATTTCCCGACGACTTACGGCTGAGCTTCTCACAACACATGCGCTTCCTCAACGGCTCCCTCTAGGTCCATCCTGGGGGGGTCGCCGGCCAATCG
3880 3890 3900 3910 3920 3930 3940 3950 3960 3970 3980 3990 400
GCAGACGGGACAGGGCCTTTTTTTTTTTTTTTTTTTGAGAGTTTTATTGACAGGTGCCAGGGCCCCGTCGACACCGATGAACTCATCGAGATCGAACAGCTCTGCCTCAGCCACCCGGCCGTCAA
CGTCTGCCCTGTCCCGGAAAAAAAAAAAAAAAAAAAAACTCTCAAAATAACTGTCCACGGTCCCGGGGCAGCTGTGGCTACTTGAGTAGCTCTAGCTTGTCGAGACGGAGTCGGTGGGCCGGCAGTT
4010 4020 4030 4040 4050 4060 4070 4080 4090 4100 4110 4120
AAAGGAGATCGCTAAGCTTAAGCTTCCACCTGGTGTGACTGTCTGCAACGATCCGTGG
TTTCCTCTAGCGATTCGAATTCGAAGGTGGACCACACTGACAGACGTTGCTAGGCACC
4130 4140 4150 4160 4170 4180

Appendix IV. Annotated DNA Sequence for Putative Non-Specific Annealing Site (Only partial sequence was reserved to avoid redundancy)

